

SaltStack: Grains, Pillars, Targeting and Render Systems

In this third and concluding article in the series on SaltStack, which makes software for configuration management, infrastructure automation and cloud orchestration, we look at the various components of SaltStack.



Let's start with the first component of SaltStack, called grains.

Grains

Grains is an interface that provides information specific to a minion. The information available through the grains interface is static. Grains gets loaded when the Salt minion starts. This means that the information in grains is unchanging. So grains information could be about the running kernel or the operating system. It is case insensitive; the names *FOO* and *foo* target the same grain.

Listing grains

Available grains can be listed by using the *grains.ls* module:

```
salt '*' grain.ls
```

Grains data can be listed by using the *grains.items* module:

```
salt '*' grains.items
```

Using grains in the *cmd* line

1. To ping all machines with the Red Hat OS, use the following command:

```
salt -G 'os:redhat' test.ping
```

2. To pull the number of cores on all machines that have a 64-bit CPU, use the command given below:

```
salt -G 'cpuarch:x86_64' grains.item num_cpus
```

Types of grains

There are two types of grains—core grains (these come with the Salt installation) and custom grains. The latter are found in:

- */etc/salt/grains*
- */etc/salt/minion*
- Grains directory, synced to minions

You can set up custom grains using either of the following methods.

Minion (the easiest and preferred option): The configuration file, by default, is */etc/salt/minion*.

Add custom static grains under the *grains:* section.

```
grains:
  datacenter: datacenter4
  environment: test
  department: HR
  role: webserver
```

In the */etc/salt/grains* file, custom grains are configured in the same way as in the example above, but without the line with the word 'grains'.